

Amelia Eckard

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Education

University of North Carolina at Charlotte <i>M.S. in Artificial Intelligence / Early Entry Program</i>	Charlotte, NC <i>Expected May 2028</i>
University of North Carolina at Charlotte <i>B.S. in Computer Science / Concentration: AI, Robotics, & Gaming</i>	Charlotte, NC <i>Expected May 2027</i>

Experience

Research Intern <i>Stevens Institute of Technology, Dr. Jina Huh-Yoo</i>	Remote <i>June 2026 – Present</i>
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- Engineered *rq-system*/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.
- Implemented dataset tagging, a seven-category research-question taxonomy, explicit screening/classification/extraction gates, and gold-sample validation using precision, recall, F1, accuracy, and confusion matrices.

Undergraduate Researcher <i>UNC Charlotte, Dr. Todd Dobbs</i>	Charlotte, NC <i>August 2025 – May 2026</i>
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- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues.
- Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.

Projects

CNN Scene Classification <i>Python, PyTorch, torchvision, CUDA</i>	2026
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- Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images.
- Improved validation accuracy from 42.25% to 51.46% through image augmentation while comparing preprocessing choices, training behavior, and model performance.

Emergency Fund Predictor <i>Python, scikit-learn, pandas, SHAP</i>	2025
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- Built an 11-feature machine-learning pipeline using 12,295 Federal Reserve SHED respondents enriched with BEA indicators to study emergency-fund vulnerability.

Resolve – LPL Financial University Hackathon <i>AI/ML, Workflow Intelligence, FinTech</i>	2026
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- Developing a proactive exception-intelligence layer that predicts likely blockers before financial-service requests visibly stall.

Technical Skills

Research/AI PyTorch, scikit-learn, NumPy, pandas, OpenCV, SHAP, Core ML, LLM APIs

Methods Precision, recall, F1, accuracy, confusion matrices, structured extraction, taxonomy design

Tools/APIs Python, Git, CUDA, PubMed E-utilities, CrossRef, Perplexity API

Certifications & Involvement

UR2PhD Training Course, Computing Research Association	December 2025
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Social and Behavioral Research, CITI Program	September 2025
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